

GRU (Gated Recurrent Unit)

What is GRU?

GRU (Gated Recurrent Unit) is a type of Recurrent Neural Network (RNN) used for processing sequential data such as Text, Speech Recognition, Time Series Forecasting, Chatbots, and Machine Translation. GRU was introduced to solve the vanishing gradient problem of traditional RNNs and is a simpler alternative to LSTM (Long Short-Term Memory).

Why do we need GRU?

Simple RNNs struggle to remember information from earlier time steps when sequences become long. GRU helps retain important information for longer periods.

Structure of GRU

A GRU has two gates: 1. Update Gate (z): Decides how much old information to keep and how much new information to add. 2. Reset Gate (r): Decides how much past information to forget.

Working of GRU

Previous Hidden State ($h(t-1)$) → Reset Gate → Candidate State → Update Gate → New Hidden State ($h(t)$)

GRU Equations

Update Gate: $z(t) = \sigma(W_z[h(t-1), x(t)])$

Reset Gate: $r(t) = \sigma(W_r[h(t-1), x(t)])$

Candidate Hidden State: $\tilde{h}(t) = \tanh(W_h[r(t) * h(t-1), x(t)])$

Final Hidden State: $h(t) = (1 - z(t)) * h(t-1) + z(t) * \tilde{h}(t)$

GRU vs LSTM

GRU: 2 gates, fewer parameters, faster training, less memory usage.

LSTM: 3 gates, memory cell, more parameters, slightly better for very long sequences.

Advantages of GRU

- Faster training
- Fewer parameters
- Less memory consumption
- Handles vanishing gradients better than RNN
- Good performance on NLP tasks

TensorFlow/Keras Example

```
from tensorflow.keras.models import Sequential
from tensorflow.keras.layers import GRU, Dense
```

```
model = Sequential([
```

```
GRU(64, input_shape=(100,10)),  
Dense(1)  
)
```

Exam Definition

GRU (Gated Recurrent Unit) is an advanced RNN architecture that uses Update Gate and Reset Gate to control the flow of information, allowing the network to remember important information for long sequences while reducing the vanishing gradient problem.